

Mi Problems: Complex Numbers

Answers to Mi problems are on page M-14. **Updated Saturday February 14, 2026 2:47pm.**

Mi1. Find the real and imaginary parts of $4i - 3$.

Mi2. Find the magnitude (aka “modulus” or “absolute value”) of $4 - 3i$.

Mi3. Find the phase (aka “argument”) of $4 + 3i$.

Mi4. Find the phase of (a) $-4 + 3i$ and (b) $-4 - 3i$. (Hint: your calculator lies. Graph it.)

Mi5. Find the real and imaginary parts of $6e^{i\pi/6}$.

Mi6. Find the magnitude and phase of $6e^{i(4\pi/7)}$.

Mi7. Find the real and imaginary parts of $(3 + 2i)(-4 - 6i)$.

Mi8. Find the real and imaginary parts of $(2+3i)/(4-5i)$. (Hint: multiply both the numerator and denominator by the denominator’s complex conjugate.)

Mi9. Plot each of the following on the complex plane: $e^{i \cdot 0}$, $e^{i\pi/2}$, $e^{i\pi}$, and $e^{i3\pi/2}$.

Mi10. Write $4 + 3i$ in the form $re^{i\theta}$.

Mi11. What are the magnitude and phase of $e^{i\pi/6}e^{i\pi/4}$?

Mi12. What are the magnitude and phase of $(4 + 3i)e^{0.2i}$?

Mi13. Suppose $f(t) = 20e^{i\omega t}$, where $\omega = 5 \text{ sec}^{-1}$. Find the magnitude and phase of $f(t)$ at (a) $t = 0$ seconds and (b) at $t = 0.27$ seconds.

Mi14. Suppose $f(t) = (6+8i)e^{i\omega t}$, where $\omega = 5 \text{ sec}^{-1}$. Find the magnitude and phase of $f(t)$ at (a) $t = 0$ seconds and (b) at $t = 0.27$ seconds.

Mi15. The “exponential rule” for derivatives of exponential functions is that $\frac{d}{dx}e^u = e^u \frac{du}{dx}$. Show by direct substitution of the Euler formula, $e^{i\theta} = \cos \theta + i \sin \theta$, that the exponential rule yields the correct results for $\frac{d}{dt}(Ae^{i\omega t})$ and $\frac{d}{dt}(Ae^{-i\omega t})$.

Answers to Selected Mi Problems:

Mi1. 4 and -3 , respectively

Mi2. 5

Mi3. 0.643

Mi4. (a) 2.50 (b) 3.78 or -2.50

Mi5. 5.20 and 3, respectively

Mi6. 6 and $4\pi/7$

Mi7. 0 and -26

Mi8. -0.1707 and 0.537

Mi9. Your points are the same as $1, i, -1$, and $-i$.

Mi10. $5e^{0.643i}$

Mi11. 1 and $5\pi/12$

Mi12. 5 and 0.843

Mi13. (a) 20 and 0 (b) 20 and 1.35

Mi14. (a) 10 and 0.927 (b) 10 and 2.277